

Starbucks Rewards Offer Response Modeling

Official Final Report

1. Project Goal

This project builds an end-to-end machine learning pipeline for the Starbucks Rewards dataset. The central task is to predict whether a customer will truly respond to a promotional offer.

We define true response carefully: the customer must receive the offer, view the offer, and complete the offer before it expires.

This definition matters because a completed offer without a prior view is closer to an organic purchase than a real offer-driven response. Keeping those cases negative prevents target leakage and keeps the problem aligned with business value.

2. Data Sources

The raw dataset has three files:

- portfolio.json: offer metadata such as offer type, reward, difficulty, duration, and channels
- profile.json: customer demographics and membership start date
- transcript.json: time-ordered customer events, including offer received, offer viewed, offer completed, and transactions

Raw JSON files are parsed into structured parquet tables under data/processed/.

3. Pipeline Summary

The final pipeline runs from raw JSON to model outputs:

1. Parse raw data
2. Build response target table
3. Merge customer and offer metadata
4. Build demographic, behavioral, and offer-level features
5. Assemble final feature matrix
6. Train models
7. Export evaluation artifacts
8. Run explainability, clustering, uplift, and budget simulation

The final modeling matrix is stored in: data/processed/features.parquet

4. Target Construction

Target logic follows four rules:

- only BOGO and discount offers are used for classification
- informational offers are excluded
- positive label requires received -> viewed -> completed
- completion must happen inside the offer duration window

The target build also stores audit fields, including completed_any, so non-view completions remain visible for analysis without contaminating the label.

5. Feature Engineering

Three feature groups feed the models.

Demographic features:

- age
- income
- gender indicators
- membership duration

Behavioral features:

- number of transactions before receipt
- total spend before receipt
- average spend before receipt
- time since last transaction
- prior offers received before current receipt
- prior offers viewed before current receipt
- prior offers completed before current receipt
- prior view and completion rates

Offer features:

- BOGO vs discount indicators
- difficulty
- reward
- duration
- web, email, mobile, and social channel indicators
- reward-to-difficulty ratio

All behavioral features are computed strictly from events that happened before the current offer receipt time.

This was a critical engineering requirement and a major review point in the project.

6. Train / Validation / Test Strategy

The final version uses a chronological split because the dataset has enough time depth.

- train cutoff: hour 336
- validation cutoff: hour 408
- test split: all later receipts

Split sizes:

- train rows: 30,420
- validation rows: 10,210
- test rows: 20,412

This is better than a random split because it respects event order and better matches real deployment.

7. Models Trained

Three supervised models were trained:

- Logistic Regression
- Random Forest
- XGBoost

Two baselines were also scored:

- majority-class baseline
- simple rule baseline

Saved models:

- models/logistic_regression.joblib
- models/random_forest.joblib
- models/xgboost.joblib

8. Evaluation Outputs

Validation and held-out test artifacts were exported as:

- ROC curve
- Precision-Recall curve
- confusion matrices
- classification report CSV
- model comparison table

Main test-set metrics:

Majority Class:

- ROC-AUC 0.500
- AP 0.323
- F1 0.000

Rule Baseline:

- ROC-AUC 0.500
- AP 0.323
- F1 0.000

Logistic Regression:

- ROC-AUC 0.746
- AP 0.538
- F1 0.576
- Precision 0.515
- Recall 0.655

Random Forest:

- ROC-AUC 0.767
- AP 0.558
- F1 0.547
- Precision 0.564
- Recall 0.530

XGBoost:

- ROC-AUC 0.785
- AP 0.586
- F1 0.573
- Precision 0.581
- Recall 0.566

Best overall model:

XGBoost

Interpretation:

XGBoost achieved the best ranking quality and best precision-recall tradeoff overall.

Logistic Regression remained competitive and produced the strongest recall, making it a good transparent benchmark.

9. Explainability

Explainability outputs include:

- logistic coefficient plot
- random forest feature importance plot
- XGBoost feature importance plot
- SHAP summary plot for the best tree model

These artifacts were exported under:

- reports/feature_importance/
- reports/figures/

10. Clustering

Customer clustering was run with K-Means using a reduced behavioral-demographic feature space.

Model selection used both elbow and silhouette diagnostics.

Best cluster count:

- $K = 4$

Cluster summary showed:

- a large low-response segment
- stronger response segments with higher prior engagement
- a small but very high-value segment with high spend and high response

Completion rate by segment and offer type was exported, which makes the clustering useful for business targeting instead of just exploratory visualization.

11. Uplift Modeling

An optional but fully working uplift module was added.

Approach:

- treatment: BOGO and discount offers
- control: informational offers
- outcome: whether a transaction occurred inside the offer window
- learner: T-learner with separate treated and control models

Main uplift results:

- treated model ROC-AUC: 0.7887
- control model ROC-AUC: 0.7609
- Qini AUC: 53.49
- top 30 percent incremental gain: 242

Interpretation:

The uplift ranking captures incremental response signal beyond simple purchase prediction.

12. Budget Simulation

A greedy budget simulation was built on top of uplift scores.

Budget result:

- budget: 13627.75
- greedy selected rows: 3119
- greedy expected incremental purchases: 124.21
- random baseline mean expected incremental purchases: 51.17

Interpretation:

Targeting by uplift score produced much higher expected incremental response than random allocation.

13. Key Observations

1. Leakage-safe target logic materially matters.
The reviewed version fixed prior-offer history leakage and kept completed-without-view cases out of the positive class.
2. Behavioral features carry the strongest signal.
Prior spend, recency, and prior offer engagement are more useful than demographics alone.
3. XGBoost is the strongest final classifier.
It performed best on both ROC-AUC and average precision on held-out test data.
4. Customer segments are materially different.
Not all customers respond the same way, and weaker segments often respond better to discount offers than BOGO offers.
5. Uplift and budget simulation add business relevance.
The project moves beyond prediction into decision support.

14. Deliverables Produced

Data:

- cleaned processed tables
- final feature matrix
- customer segment table
- uplift feature table

Models:

- Logistic Regression
- Random Forest
- XGBoost
- uplift treatment model
- uplift control model

Reports:

- validation metrics
- test metrics
- model comparison table
- clustering summaries
- uplift metrics
- budget simulation summary

Figures:

- ROC
- PR
- confusion matrices
- EDA charts
- feature importance plots
- SHAP summary
- clustering diagnostics
- clustering heatmaps
- uplift charts
- budget comparison chart

15. Final Conclusion

The project is complete as a strong, submission-ready ML system.

It now includes:

- correct target construction

- leakage-safe feature engineering
- reproducible model training
- held-out evaluation
- explainability
- clustering
- uplift modeling
- budget simulation

The final system demonstrates both technical rigor and clear business usefulness .